

Question	Marking Guidance	Mark	Comments
06.1	GgX ^R X ^r ;	1	1. Accept alleles in any order. 1. Accept GgRr with alleles in any order.
06.2	If it were recessive all flies of 3 and 4 would be grey OR 3 and 4 produce 9/black (fly) OR Grey parents produce black (fly);	1	
06.3	1. (Fly) 3 (and 4) produce 9/black (fly) OR (Fly) 9 would not be black OR (Fly) 9 would be grey OR Grey parents/male produce black female (fly); 2. (Fly) 3 would pass dominant <u>allele</u> to 9; 3.(Fly) 2 (and 1) produce 5/grey (fly) OR Black female produces grey male OR (Fly) 5 could not be grey OR (Fly) 5 would be black; 4.(Fly) 5 would receive recessive <u>allele</u> from 2;	2 max	Mark in pairs 1 and 2 or 3 and 4. 2. Accept <u>allele</u> for grey colour would be passed on by 3. 4. Accept <u>allele</u> for black colour would be passed on by 2.
06.4	1. GgX ^r X ^r and ggX ^R Y; 2. GgX ^R X ^r , ggX ^R X ^r , GgX ^r Y and ggX ^r Y; 3. Grey-bodied red-eyed female, black-bodied red-eyed female, grey-bodied white-eyed male, black-bodied white-eyed male and ratio 1 : 1 : 1 : 1;	3	1 and 2. Accept the following alternative notations for sex-linked crosses e.g. for mp 1 Ggrr and ggRY or Ggrr and gg R- or Ggrr and ggR i.e. space for Y

	If 1, 2 and 3 incorrect allow one mark for correct gametes from incorrect dihybrid parental genotypes.		chromosome. 2 and 3. Accept any order of genotypes and phenotypes. 3. Accept sequence of phenotypes does not need to mirror genotypes but must be correct. 3. Accept alternative ratios in correct proportions e.g. 4:4:4:4
06.5	1. Correct answer of 48% = 2 marks;; 2. Incorrect answer but shows understanding that $2pq$ = heterozygous/carriers = 1 mark OR Incorrect answer but shows understanding that $1 - (p^2 + q^2)$ = heterozygous/carriers = 1 mark;	2	1. Accept 0.48 for 1 mark. 2. Accept understanding of $2pq$ by using a calculation involving 2 x two different numbers.
